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19 July 1985
SUPERSEDING
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(See 6.6)

MILITARY SPECIFICATION

MITTENS, DIVERS FOR MK-12 SURFACE SUPPORTED DIVING SYSTEM

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the MK-12 diver's mittens.

1.2 Classification. The mitten shall be of one type in the following sizes (see 6.2.1):

Size 1 - Small
Size 2 - Medium
Size 3 - Large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4220

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SPECIFICATIONS

FEDERAL

A-A-1683 - Tape, Pressure-Sensitive Adhesive (Packaging, Paper).
DDD-L-20 - Label: For Clothing, Equipage, and Tentage (General Use).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 395 - Rubber Property-Compression Set, Standard Test Method for. (DoD adopted)
D 412 - Rubber Properties in Tension, Standard Test Method for. (DoD adopted)
D 1056 - Flexible Cellular Materials-Sponge or Expanded Rubber, Standard Specification for. (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials.

3.2.1 Chloroprene foam. The chloroprene foam for the mittens shall be pigmented black, gas expanded chlorinated cellular rubber, having a closed cell structure conforming to RE-41-E1 of ASTM D 1056 and as specified herein when tested as specified in 4.6.1. The chloroprene foam shall have a minimum density of 10 and a maximum density of 20 pounds per cubic foot (lb/ft^3). The foam thicknesses used for the mittens shall be 1/4 $\pm$ 1/16 inch thick with no skin on both sides; 1/4 $\pm$ 1/16 inch thick with skin on one side only. The foam material shall conform to table I when tested as specified in 4.6.1. A certificate of compliance shall be required to show the chloroprene foam meets the requirements of table I (see 6.2.2).

TABLE I. Foam.

Characteristic	Requirement
Compression set under constant compression	25 percent maximum
Compression deflection	2.0 lb/in^2 minimum 7.0 lb/in^2 maximum
Water absorption	6.0 percent maximum
Accelerated aging shrinkage	5.0 percent maximum in each direction
Tensile stress	135 lb/in^2 maximum (500 percent elongation)

3.2.2 Knitted stretch cloth. The stretch cloth shall be circular knit from 100 percent continuous filament nylon yarn and conform to the physical requirements of table II when tested as specified in 4.6.1. A certificate of compliance shall be required to show the knitted stretch cloth is of the specified material (see 6.2.2).

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TABLE II. Knitted stretch cloth.

Knitted stitch	Yarn per inch (minimum) wales courses	Weight, ounce per sq yd	Elongation, percent wales courses	Color
Jersey	36 by 38	2.4 ± 0.2	150 minimum 225 minimum 225 maximum 400 maximum	Black (outside) Blue (inside)

3.2.2.1 Colorfastness. The knitted cloth shall show color fastness to laundering, crocking, and perspiration equal to or better than the standard sample when tested as specified in FED-STD-191. When no standard sample is available the finished material shall show a minimum of "good" fastness to laundering, perspiration and a Munsell value for dry and wet crocking not lower than 8.5. Testing shall be as specified in 4.6.1.

3.2.3 Lamination. The stretch cloth shall be laminated to the different foam thicknesses as follows:

(a) 1/4 inch thick foam, which has two cut surfaces, stretch cloth laminated to both sides.
(b) 1/4 inch thick foam, which has one cut surface and one skin surface, stretch cloth laminated to cut surface side.

The cloth shall be laminated to the chloroprene foam by means of a suitable adhesive, or other process capable of meeting the physical requirements of the laminated foam material. The adhesive used shall be non-toxic, non-irritant and free from objectionable odor. The fabric backed (laminated) side shall be smooth and free of wrinkles or puckers, and show no signs of blocking, tackiness or adhesive strike through when tested in 4.6.1.

3.2.3.1 Laminated foam. The finished laminated foam material shall conform to table III when tested as specified in 4.6.1.

TABLE III. Laminated foam.

Characteristics	Requirements
Laundering resistance	No delaminations
Adhesion, original	8.0 pounds minimum per 2 inches
Adhesion, after laundering	8.0 pounds minimum per 2 inches
Adhesion, after aging	8.0 pounds minimum per 2 inches
Blocking	No blocking

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3.2.4 Adhesive cement. The cement used to seal the seams and reinforcement pieces shall be a chloroprene compound. A certificate of compliance shall be required to show the adhesive cement is the specified compound (see 6.2.2).

3.2.5 Cemented seams. The abutted and cemented seams (without reinforcement piece) shall have an initial and after aging minimum breaking strength of 35 pounds per 2 inches when tested as specified in 4.6.1.

3.2.6 Label. The label shall conform to type I, class 1 of DDD-L-20. The label shall be applied to the inside of mitten centered plus or minus 1/4 inch off center tolerance on back hand portion of gauntlet; 1 ± 1/8 inch from top edge of the mitten. The inscription shall read as follows:

"Mitten, Divers MK-12
Stock No: 4220-00-000-0000 (Example)
Contract No: DLA 100-00-0-0000 (Example)
Name of Supplier:"

3.2.6.1 Size identification. Each mitten shall have a size identification placed on the outside of the back hand portion of gauntlet as shown on figure 1. The material for the size identification shall conform to 3.2.1 having yellow nylon - one on face side and black nylon one on back side. The thickness of the foam shall be 1/8 inch. The size identification shall be approximately 1 inch in height.

3.2.7 Strapping. The material for the strapping, which is placed between the two fingers of the mittens shall be a strip of laminated foam, one side, 1/8 inch thick. The strip of strapping shall be approximately 3/8 inch wide and have a length which extends approximately 1/2 inch on either side of seam. A certificate of compliance shall be furnished for the strapping material as specified in 4.1 (see 6.2.2).

3.2.8 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Design. The mitten shall be a trigger finger design and made of laminated chloroprene foam material. Seams shall be cemented as shown on figure 1.

3.4 Pattern parts. The component parts of the mitten shall be cut from the material specified in accordance with table IV.

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TABLE IV. Pattern parts.

Material	Nomenclature of pattern parts	Cut parts
Laminated foam both sides, 3/4 inch thick	Hand (top and bottom) Thumb	2 2
Laminated foam one side, 1/4 inch thick	Gauntlet	2
Laminated foam both sides, 1/8 inch thick	Size numbers 1 2 3	 2 2 2

3.5 Manufacturing requirements. The mittens shall be manufactured in accordance with the operation requirements specified in 3.5.1 through 3.5.11. The contractor is not required to follow the exact sequence of operations listed, provided the finished mittens are identical to that produced by the following sequence of operations.

3.5.1 Cutting. The component parts of mittens shall be cut in accordance with the patterns furnished specifying size and shape for proper assembly of all component parts.

3.5.2 Replacement of defective components. Components classified as defective during the spreading, cutting, and manufacturing process shall be removed from production and replaced with nondefective and properly matched components.

3.5.3 Procedure for seam cementing. The procedure for seam cementing shall be as follows:

(a) Check surface of the seams ensuring they are clean, dry, and free from grease and oil.
(b) Each surface shall be given two coats of cement (pre-cementing) allowing 10 minutes drying time between coats. Check the surfaces to ensure the pre-cementing procedure has completely covered the cellular structure of the foam.
(c) The third and final coat of cement shall completely cover the previous two coats. Allow the coat to become tacky and press the cemented edges together firmly. Turn to inside and press the back side of the seam together. After pressing the seams inside and outside, the seams shall be pinched together inside and outside.

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3.5.4 Procedure for strapping piece cementing. The procedure for strapping piece cementing shall be as follows:

(a) Check surfaces ensuring they are clean, dry and free from grease and oil.
(b) The nylon surfaces of the mitten shall be given three pre-coats of cement. The strapping shall be given one pre-coat of cement applied onto the open-cell foam side of the reinforcement piece. Allow 10 minutes of drying time between coats.
(c) After the pre-cementing coats have dried, apply a final coat to each surface. When tacky, the surfaces may be joined by rolling with a heavy metal roller.
(d) After allowing sufficient time for the bond between the two surfaces to set, the edges of the strapping piece shall be rolled down. To achieve this, pre-cement the edges of the strapping piece and the nylon of the mitten where there will be contact with the edges. When dry, cement the surfaces and roll the edges of the strapping piece over onto the surfaces of the mitten (see figure 2).

3.5.5 Make thumb. Apply cement to the thumb joining seams, abut and press firmly together accordingly to the procedure specified in 3.5.3.

3.5.6 Attach thumb to thumb hole in hand. Apply cement to the thumb and thumb hole joining seams. Align thumb to thumb hole by matching guide in the thumb hole with the thumb joining seam. Abut and press firmly together as specified in 3.5.3.

3.5.7 Close hand. Apply cement to the hand joining seams, abut and press firmly together, matching guides according to the procedure specified in 3.5.3.

3.5.8 Make gauntlet. Apply cement to the gauntlet joining seams, abut and press firmly together, as specified in 3.5.3 The jersey side of the gauntlet shall finish on the outside.

3.5.9 Attach gauntlet to hand. Apply cement to gauntlet and hand joining seams. Align the seams of the gauntlet and hand, abut and press firmly together according to the procedure specified in 3.5.3.

3.5.10 Apply strapping piece. Apply a strip of strapping between the two fingers of the assembled hand according to the procedure specified in 3.5.4.

3.5.11 Label and size identification. The label and size identification shall be applied to the mittens as specified in 3.2.6 and 3.2.6.1 with adhesive specified in 3.2.4.

3.6 Sizes and measurements. The finished measurements of the mittens shall be as shown in table V. Measurements and tolerances in table V are expressed in inches.

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TABLE V. Finished measurements.1/

Size	(A) Length	(B) Finger length	(C) Mitten finger length	(D) Thumb length	(E) Width (palm)	(F) Opening
1 Small	14-5/8	2-7/8	3-3/8	2-3/4	4-1/4	3-7/8
2 Medium	14-7/8	3.0	3-1/2	2-7/8	4-1/2	4-1/4
3 Large	15-1/8	3-1/8	3-5/8	3.0	4-3/4	4-5/8
Tolerance	±1/4	± 1/8	± 1/8	± 1/8	± 1/4	± 1/4

1/ Measurements shall be taken with the mitten laid flat, palm facing up except when otherwise specified. Refer to figure 1 for A, B, C, D, E, and F measurements.

3.7 Workmanship. The finished mittens shall conform to the quality of product established by this specification and the occurence of defects shall not exceed the applicable acceptable quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article submitted in accordance with 3.1 shall be inspected as specified in 4.7.1 for compliance with design, construction, workmanship and dimensional requirements.

4.4 Quality conformance inspection.

4.4.1 Sampling for quality conformance inspection. Sampling and inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein. For purpose of sampling, an inspection lot for quality conformance inspection shall consist of all material of the same type offered for delivery at one time. Lot and sample size shall be as specified in table VI.

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TABLE VI. Sample size.

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.2 Lot. The lot shall be unacceptable if one or more units fail to meet any requirements specified. The basis for lot size for laminated and unlaminated chloroprene foam shall be one sheet and 1 yard full width for the knitted stretch fabric. The sample unit for testing shall be as follows:

(a) Unlaminated foam - 1 square yard.
(b) Laminated foam - 1 square yard.
(c) Knitted stretch fabric - 1 yard full width.

4.5 Examination.

4.5.1 Visual examination. Visual examination shall be as specified in table VII.

TABLE VII. Visual defects.

Defect	Classification	
	Major	Minor
MATERIAL DEFECTS AND WORKMANSHIP DAMAGES		
(a) Any hole.	X	
(b) Deep creases or fold lines.	X	
(c) Cut or tear.	X	
BASIC MATERIAL		
(a) Any evidence of delamination.	X	
(b) Area of excess cement, for example solid runs, ridges.		X
(c) Chloroprene foam or cement tacky.	X	
(d) Any brittle area which cracks when bent.	X	
(e) Any imbedded foreign material which when removed results in damage to basic material.	X	
(f) Any imbedded foreign material which when removed, does not damage the basic material.		X
(g) Any blister or pin hole.	X	
(h) Any objectionable odor.		X
(i) Color not as specified.		X

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TABLE VII. Visual defects. - Continued

Defect	Classification	
	Major	Minor
CLEANNESS		
(a) One or more spots or stains of a permanent nature.		X
(b) Unsightly cement mass.		X
COMPONENTS AND ASSEMBLY		
Any component part omitted or any required operation omitted or improperly performed (unless otherwise classified herein).	X	
CUTTING		
Any component part not cut in accordance with marks indicated on pattern or not in accordance with specification requirements.	X	
CEMENTED SEAMS		
(a) Any open seam.	X	
(b) Twisted or puckered seam.	X	
(c) Seam not properly joined.	X	
REINFORCEMENT PIECE (STRAPPING)		
(a) Reinforcement piece, omitted; not positioned as specified.	X	
(b) Reinforcement piece insecurely cemented.	X	
GAUNTLET		
(a) Insecurely attached to mitten.	X	
(b) Assembled with skin side on outside.	X	
(c) Ends of gauntlet when assembled, finishing uneven with each other.	X	

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TABLE VII. Visual defects. - Continued

Defect	Classification	
	Major	Minor
THUMB		
(a) Insecurely attached to thumb hole in mitten marking.	X	
(b) Label omitted, illegible or incorrect.	X	
(c) Label not positioned and centered (1/4 inch off center tolerance) on inside of back hand portion of gauntlet.		X
(d) Label measured from top edge of gauntlet less than 7/8 inch or more than 1-1/8 inches.		X
(e) Size identification omitted or not positioned as specified.		X
PAIRING		
(a) Any pair mismatched, for example, two lefts or two rights.	X	

4.5.2 Dimensional examination. The mittens shall be examined for defects in dimensions. Any measurement deviating from nominal dimensions and tolerances specified shall be scored as a measurement defect.

4.5.3 Inspection levels and acceptable quality levels. The inspection levels and acceptable quality levels (AQL's) expressed in defects per 100 units, for visual and dimensional examination shall be as follows:

	AQL's	Inspection level
For examination in 4.5.1		
Major	2.5	II
Total (major and minor combined)	6.5	II
For examination in 4.5.2		
One class	4.0	S-3

4.6 Tests.

4.6.1 Testing of components. In addition to the quality assurance provisions of the referenced specification, components specified in table VIII shall be tested for the characteristics shown therein.

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TABLE VIII. Testing of components.

Components	Characteristic	Determinations	Requirement paragraph	Test methods
Chloroprene foam	Material identification	-		
	Color	1	3.2.1	4.5.2
	Thickness	5	3.2.1	ASTM D 1056
	Compression set under constant deflection	5	3.2.1	4.6.1.1
	Compression deflection	5	3.2.1	ASTM D 1056
	Shrinkage after accelerated aging	3	3.2.1	4.6.1.2
	Water absorption	3	3.2.1	ASTM D 1056
	Density	5	3.2.1	4.6.1.3
	Tensile stress	5	3.2.1	4.6.1.5
Knitted Nylon Stretch Cloth	Material identification	-	3.2.2	
	Weight	5	3.2.2	FED-STD-191, method 5040
	Wales and courses	5	3.2.2	FED-STD-191, method 5070
	Elongation	5	3.2.2	4.6.1.4
	Color	-	3.2.2	4.5.2
	Colorfastness:			
	Laundering	1	3.2.2.1	FED-STD-191, method 5614
	Crocking	1	3.2.2.1	FED-STD-191, method 5651
	Perspiration	1	3.2.2.1	FED-STD-191, method 5680
Laminated	Adhesive strike through	-	3.2.3	4.5.2
Foam	Wrinkled or puckered	-	3.2.3	4.5.2
	Laundering resistance	5	3.2.3.1	FED-STD-191, method 5556 1/
	Adhesion:			
	Initial	5	3.2.3.1	FED-STD-191, method 5950 2/
	After 3 launderings	5	3.2.3.1	FED-STD-191, methods 5556 and 5950 1/, 2/
	After aging	5	3.2.3.1	4.6.1.2 and FED-STD-191, method 5950 2/

See footnotes at end of table.

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TABLE VIII. Testing of components. - Continued

Components	Characteristic	Determinations	Requirement paragraph	Test methods
Laminated foam	Blocking	3	3.2.3.1	FED-STD-191, method 5872 3/
Adhesive cement	Material identification	-	3.2.4	
Cemented seam	Breaking strength: Initial	5	3.2.5	FED-STD-191, method 5100 4/, 5/
	After aging	5	3.2.5	FED-STD-191, method 5100 4/, 5/, 6/

1/ Laundering. Three cotton laundering cycles shall be performed on 8- by 8-inch specimens, using 1/4 inch double laminated foam.
2/ Adhesion. The distance between the clamps shall be 2 inches at the start of the test. Slitting the specimen between the nylon knit and foam to help separate the fabric from the foam sufficiently to fit the clamps is permissible. The adhesion shall be tested in the fabric's length (wales) direction and on one side only of the 1/4 inch foam laminated on both sides.
3/ Blocking. Except specimens 4 by 4 inches shall be placed together fabric side to fabric side.
4/ Cemented seams shall be centered equidistant from the clamps. Breaks shall be made with the seam at right angles to load direction. The clamps shall be 1 inch apart at the start of the test and the face of each jaw shall be not less than 1 by 2 inches. The cemented seam test specimens shall be rectangular, 2 inches wide by not less than 4 inches long.
5/ The test shall be performed on specially prepared cemented seams similar to those made on the end item.
6/ The specimen shall be suspended vertically in the air-oven at 158 ± 2 degrees Fahrenheit (°F) for 7 days. At the conclusion of the aging interval, the specimens shall be cooled at room temperature on a flat surface and allowed to rest for 4 hours before determination of adhesion.

4.6.1.1 Compression set under constant deflection. The circular die used to prepare standard test specimens shall have an inside diameter measurement of 1.129 ± 0.001 inches. Test specimens shall have a 1 square inch surface area with the top and bottom surfaces parallel to each other. Thickness of test specimens shall be 1/2 inch. If a 1/2-inch thick specimen is not readily available, then two 1/4-inch thick specimens may be plied to obtain the required 1/2 inch thickness. The apparatus and procedure shall be the same as that specified in method B of ASTM D 395. The test specimens shall be compressed to 50 percent of their measured original thicknesses. The test shall be run at room temperature for 22 hours. The load is released at the end of the test period and the thickness measured after 24 hours at room temperature. Thickness is measured as specified in ASTM D 1056. Chromium plated metal plates are not required for the apparatus. Any stiff plates that are clean and smooth may be used.

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Calculate the percentage compression set as follows:

$$\text{Compression set, percent} = \left[(t_o - t_1)/(t_o - t_s)\right] X100$$

Where:

t_o = original thickness
t_1 = thickness of specimen after specified recovery period
t_s = thickness of spacer bar used.

4.6.1.2 Shrinkage after accelerated aging. Three square specimens each with a minimum dimension of 8 by 8 inches shall be measured to the nearest 1/32 inch in both length and width direction by means of a steel rule. The specimens shall be suspended vertically in the air-oven at 158 ± 2°F for 7 days. At the conclusion of the aging interval, the specimens shall be cooled at room temperature on a flat surface and allowed to rest not less than 4 hours before determination of shrinkage. The average shrinkage in each direction shall be calculated as follows and reported to the nearest 0.1 percent:

$$\text{Percent shrinkage} = \frac{O - A}{O} \times 100$$

Where O = original length or width (inches)
A = length or width after aging (inches).

4.6.1.3 Density. Five specimens each 2 inches square shall be weighed separately to the nearest 0.001 gram on an analytical balance. The thickness of each specimen shall be determined in accordance with ASTM D 1056 and the following formula shall be used to determined density:

$$D = \frac{0.95 \times W}{T}$$

Where D = density in lb/ft^3
W = the sum of the weights of the five specimens (grams)
T = the sum of the thicknesses of the five specimens (inches).

4.6.1.4 Elongation. An incline plane type of machine (for example, I-P-4 or similar type) with a suitable autographic recording device shall be used. Specimens of the fabric shall be cut 4 by 6 inches with the long direction parallel to the direction to be tested. The specimen shall be evenly aligned without tension or slack, the 1- by 3-inch clamps set 2 inches apart. The machine shall run through two complete cycles, proceeding to its maximum depression so as to exert a 5 pound tension. Five specimens, each containing different wales and courses shall be tested, using individual recorded elongation graphs. The average stretch of five specimens for each direction shall be measured at the peak of the last recorded cycles, in a line perpendicular to the horizontal zero axis and reported to the nearest percent.

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4.6.1.5 Tensile stress. The test specimen shall be dumbbell shape and cut using die "C" as specified in ASTM D 412. Specimens shall be cut in the length direction of the sample. The tensile stress test shall be made on a power driven machine equipped with suitable jaws to firmly hold the specimen in place while they are separating at 20 ± 2 inches per minute. The machine shall be equipped with a recording device so that the applied force is accurately measured during the extension of the specimen. The test specimen shall be conditioned for 4 hours in a conditioned atmosphere of 65 percent relative humidity and 70°F. Dumbbell specimens shall be marked with a marker having a marking edge 1/16 inch in thickness. The sample shall be under no tension at the time it is marked. Marks shall be placed equidistant from its center and perpendicular to its longitudinal axis. The center of the marks shall be 1.0 ± 0.003 inch apart. Set the grip separation at 2.5 inches minimum. Place the dumbbell tabs in the grips of the testing machine, using care to adjust them symmetrically in order that the tension will be distributed uniformly over the cross section. Start the machine and note continuously the distance between the two bench marks, determining the elongation. Record the stress at 500 percent elongation. Calculate the stress by dividing the load at 500 percent elongation by the original cross-sectional area of the specimen.

$$\text{Stress } (\text{lb/in}^2) = \frac{\text{Loads (pounds) at 500 percent elongation}}{\text{Width (inches) X thickness (inches)}}$$

No single value shall be more than the maximum requirement as specified in table I.

4.6.2 Examination of the end item. Examination of the end item shall be in accordance with 4.5.1 and 4.5.2. The applicable inspection levels and AQL's shall be as specified in 4.5.1. The sample unit for the examination shall be one mitten and the selection shall be by pairs. The sample unit for pairing defects shall be one pair. Defects of pairing shall be classified as a single defect. The mittens shall be examined for defects in design, material, workmanship, and marking and the defects classified in accordance with table VIII.

4.7 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking complies with the section 5 requirements. Defects shall be scored in accordance with table VIII. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Examination for the closure defect specified in table IX shall be made on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

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TABLE IX. Packaging defects.

Examine	Defect
Marking (interior and exterior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: polyethylene bags with heatsealed seams that are crooked, non-continuous, open or not parallel. Closure of shipping container not as specified. Bulged or distorted containers.
Content	Number of mittens per package and shipping container is more or less than required.

4.7.1 Inspection of interior and exterior containers. When interior and exterior containers are required to be in accordance with PPP-B-601, examination for defects in the assembly and closure (for fiberboard containers) shall be in accordance with the appendix of the applicable container specification.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Polyethylene bag. Each pair of mittens, reversed finger tips to gauntlet, with the palms facing shall be inserted into a snug-fitting clear polyethylene film bag of 0.00125 inch thickness (plus or minus 25 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the seal made as close as possible to the open end. A 1/4-inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. As an alternate, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap, heat seal closure and corner hole not required.

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5.1.1.2 Unit package. Each pair of mittens packaged as specified in 5.1.1.1 shall be placed in a weather resistant folding paperboard box with variety, style, type, processing and class optional but conforming to PPP-B-566. The mittens shall be placed in the box reversed end to end. The inside dimensions of each paperboard box shall be approximately 15-1/2 inches in length, 5-1/4 inches in width and 3-3/4 inches in depth. Box closure shall be secured with gummed paper conforming to A-A-1683.

5.1.2 Commercial. Mittens shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2.1).

5.2.1 Level A. Twenty-four pair of mittens, of one size only, preserved as specified in 5.1, shall be packed in a close fitting box conforming to PPP-B-601, overseas type, grade B, style optional. Strapping shall be zinc coated.

5.2.2 Level B. Twenty-four pair of mittens, of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to class weather-resistant, type SF, grade V2s, style RSC of PPP-B-636. Level A packages shall be packed flat, two in length, three in width, and four in depth. Inside dimensions of each shipping container shall be approximately 31-1/2 inches in length, 16 inches in width and 15-1/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-636. Toward the end of the contract, or when there are less than the required amount per shipping container of the same size, mixed sizes may be packed within the same container.

5.2.3 Commercial. Mittens, of one size only, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required (see 6.2.1), intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129. Marking for commercial shipments shall be in accordance with ASTM D 3951.

5.3.1 Labels, mixed sizes. Each shipping container, packed with mixed sizes, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly printed or stamped thereon, and under these words shall be legibly printed or stamped the correct quantities and sizes contained therein.

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6. NOTES

6.1 Intended use. The mittens are intended to be worn as part of the MK-12 diving system which is designed to keep personnel dry in a diving situation.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) When first article is required (see 3.1).
(d) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(e) Special marking required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirement List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.2.1, 3.2.2, 3.2.4 and 3.2.7	Certificate of compliance	DI-E-2121	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

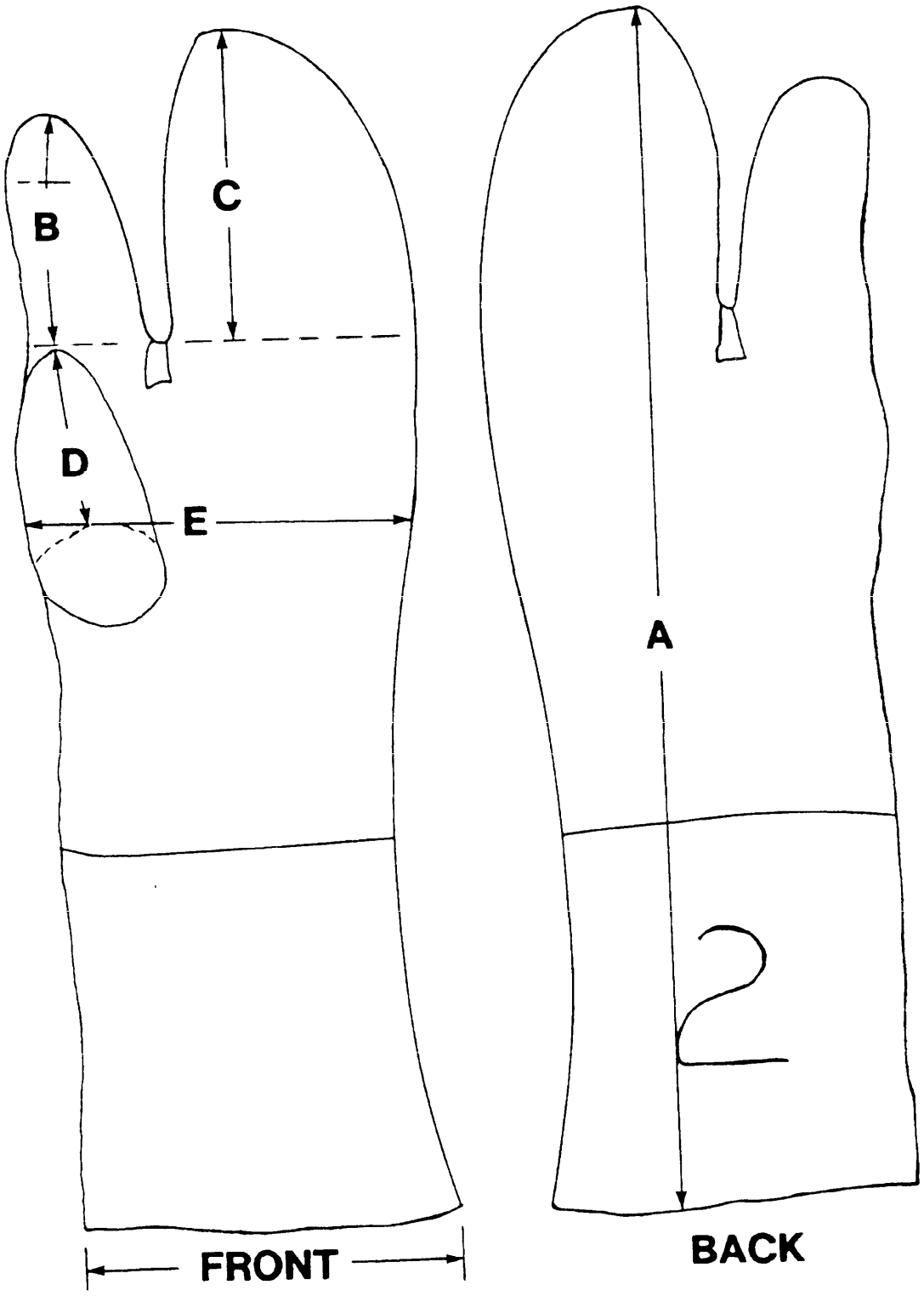
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6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4220-N303)

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SH 131925

FIGURE 1. Mittens, diver MK-12.

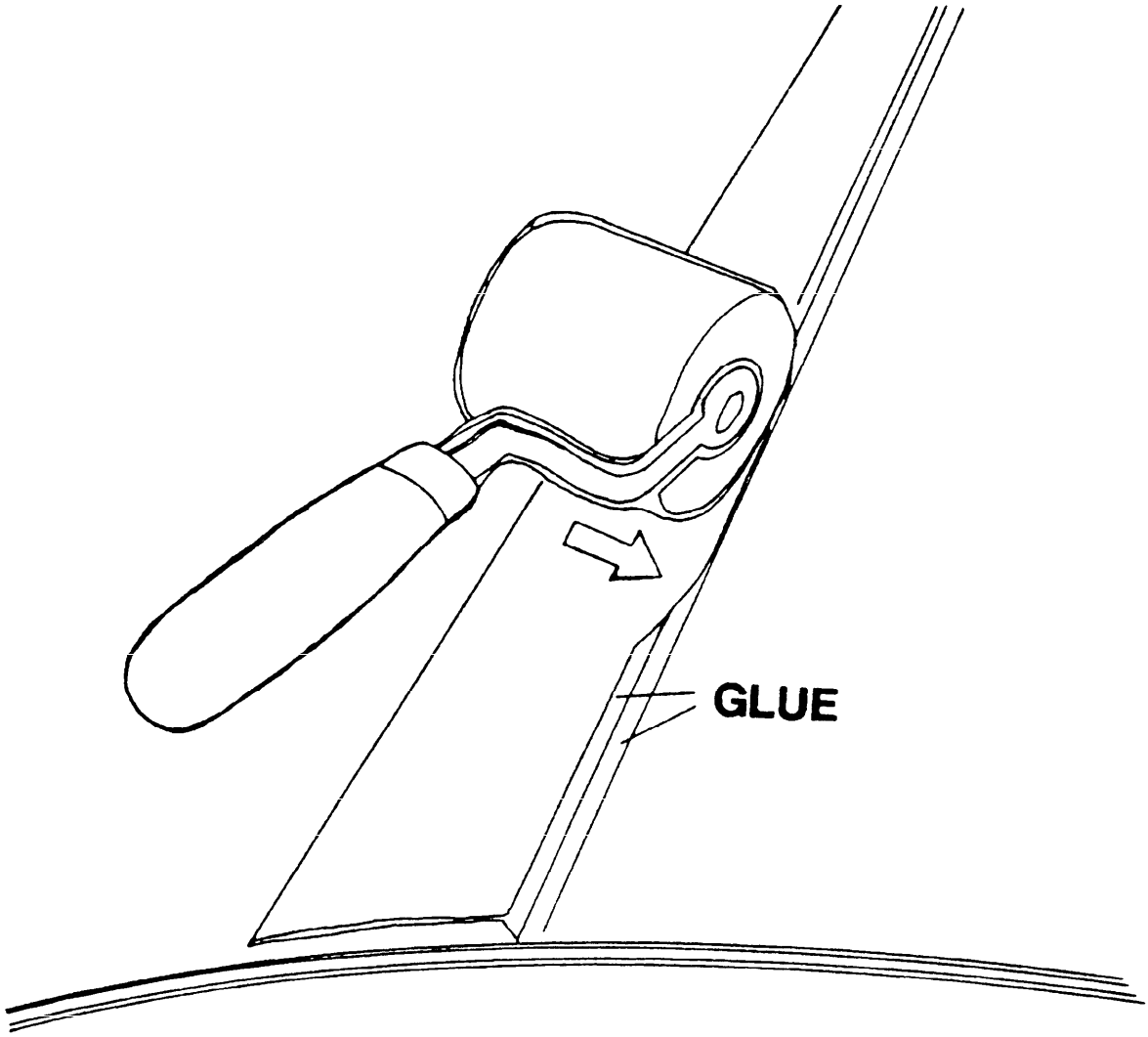
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Notes for figure 1.

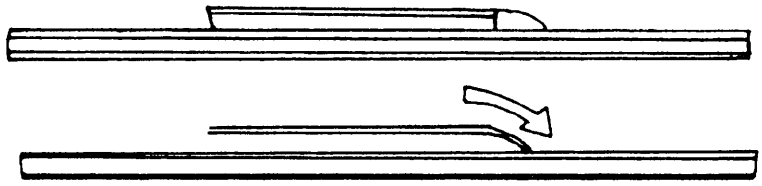
1. Length - Measurement shall be taken from end of guantlet to highest point of mitten finger on back of mitten.
2. Finger length - Measurement shall be taken from crotch of finger to the tip of finger.
3. Mitten finger length - Measurement shall be taken from crotch of mitten finger to tip of mitten finger.
4. Thumb length - Measurement shall be taken from crotch of thumb to its tip along its inner edge.
5. Width - Measurement shall be taken across the width of mitten from folded edge to folded edge in line with crotch of thumb.
6. Opening - Measurement shall be taken across the width at bottom of gauntlet.

FIGURE 1. Mittens, diver MK-12. - Continued

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SIDE VIEW FROM END



SH 131926

FIGURE 2. Strapping application.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-87029B(SH)	MITTENS, DIVERS FOR MK-12 SURFACE SUPPORTED DIVING SYSTEM

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTICE OF VALIDATION

INCH-POUND

MIL-M-87029B(SH)
NOTICE 1
10 December 1990

MILITARY SPECIFICATION

MITTENS, DIVERS FOR MK-12 SURFACE SUPPORTED DIVING SYSTEMS

MIL-M-87029B(SH), dated 19 July 1985 has been reviewed and determined to be valid for use in acquisition.

Preparing activity:
Navy - SH

AMSC N/A

FSC 4220

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